

Misaligned Voting Diagnostics ^{*}

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Abstract

Misaligned voting, when one casts a vote for a candidate they do not prefer to all others, is an important characteristic of voting systems. It may implicate strategic voting, confusion, or other behaviors. Misaligned voting is difficult to identify, and current methods may make biased estimates in identifying the overall rate and in identifying differences between voting systems. They are also ill-equipped to take in more than one dimension of voter preferences. We propose two new metrics, the convex hull and weak domination approaches, and test their performance in a series of simulations.

Keywords: misaligned voting, strategic voting, electoral systems, voting rules

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1 The Problem

We want to know how often people vote against their preferences, especially as that implicates strategic voting. If we knew who voters took to be their best vote and compared that to their actual, we could determine how many voted in a way that is misaligned with their preference. We could then learn about the circumstances under which people do so.

The current state of the art is to ask voters to rate all the parties on the ballot and compare that with to their vote. Any vote cast for the party that did not receive the highest rating is considered misaligned. A justifiable vote, by this light, is any that went to a party that had no alternatives rated higher. There are a few problems with this definition. There are multiple justifiable reasons why someone might vote a certain way, including but not limited to party preference. Variation in misaligned vote rate by voting system could be an artifact of our definition/measurement: in systems with more parties, variations in party preference are likely smaller. In systems with weak parties, individual candidates might matter more. The most important weakness of this definition is that it does not admit for multiple dimensions on which people make decisions. A good metric would scale to arbitrarily many dimensions.

2 Convex Hull

One solution is the convex hull approach (known hereafter as metric “C”). This metric defines a misaligned vote as any that falls within the convex hull created by embedding the available candidates into euclidian space on an arbitrary number of rating dimensions, given by a voter. For example, in figure 1, a voter rates candidate **a** highly on individual candidate characteristics, but rates her party relatively low. Candidate **b**, on the other hand, is rated low on individual characteristics but high on party. Depending on how highly a voter weights the importance of candidate versus party, voting for either could be justified. The line between points **a** and **b** represents a frontier at which any candidate would be a justifiable choice. Likewise, if a voter cared nothing for the individual characteristics dimension, then voting for any candidate sharing a vertical line with **b** is justified. The same reasoning applies to a voter who cares nothing for the party dimension and

the horizontal line of **a**.

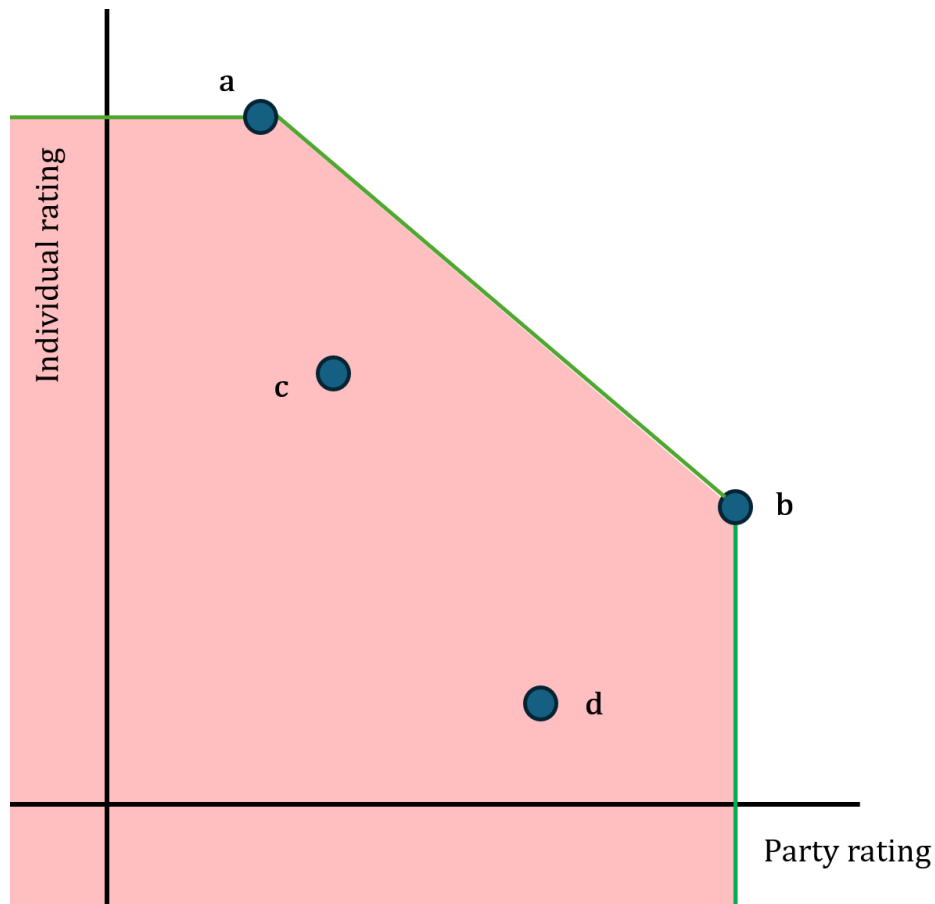


Figure 1: Convex Hull

The convex hull connecting **a**, **b**, and two points (not pictured) that share the same value in the dimension in which each performs best, but very low in all others, creates a convex hull. This construction is general to any number of dimensions and any number of candidates. Candidates **c** and **d** are both inside this convex hull. By definition, a vote for either would be a misaligned vote. No linear combination of the dimensions justifies voting for **c** or **d** over **b** or **a**. No matter the voter's weighting, there is always a better candidate. Taken farther, this metric provides a measure of magnitude: the euclidian distance from a candidate to the convex hull represents *how* misaligned a vote for that candidate is.

3 Weak Domination

While metric C defines anything inside the linear combination of the attributes of the best-performing candidates misaligned, one might consider voters with non-linear preferences. It is entirely possible that a moderate level of individual and party preference is better, in the mind of a voter, than a high level of one and low level of the other. Candidate **c** exemplifies this possibility. There is, however, a principled metric that both admits nonlinear combinations of attributes while identifying some votes as misaligned. Weak Domination (metric D) is the most conservative viable measure. It defines a vote as misaligned if it is cast in favor of a candidate who performs worse on *all* dimensions, compared to at least one other available candidate.

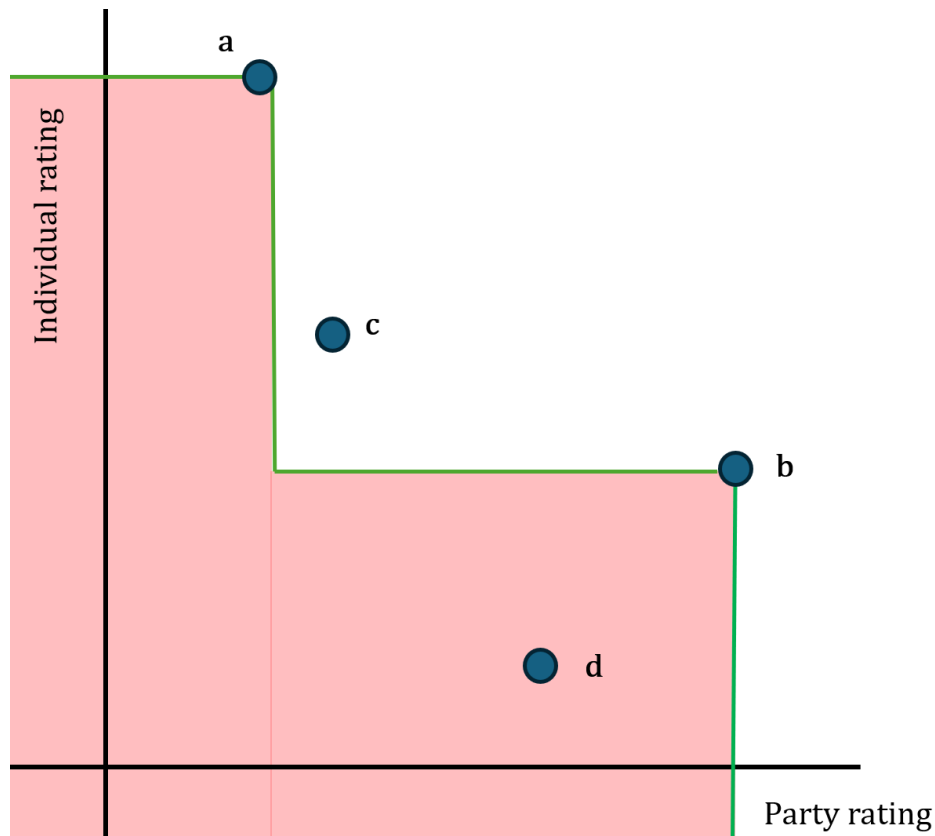


Figure 2: Weak Domination

In effect, this metric creates a series of n-dimensional rectangular prisms with a candidate at the vertex that is farthest the origin in positive directions. Any candidate that falls within another's prism is considered unjustifiable. For example, figure 2 shows that candidate **d** is worse than

candidate **b** on all dimensions. A vote for **d** would be a misaligned vote. Candidate **c**, on the other hand, is now justifiable. This metric, too, is generalizable to n dimensions and m candidates. Like metric C it also admits a distance measure – the perimeter of the union of the rectangular prisms is a frontier and a candidate’s euclidian distance to that frontier is the magnitude of misalignment for a vote cast in their favor.

4 Modeling

4.1 Baseline

Now we intend to test our metrics and other alternatives. A good metric would meet the three criteria:

1. Detect misaligned voting
2. Fail to detect differences between voting systems when none exist
3. Detect differences between voting systems when one exists

To do identify whether metrics meet these criteria, we need to know the ground truth: How often do people cast a misaligned vote? For our purposes, a misaligned vote retains its original definition: a vote cast for a candidate that is worse, from the point of view of the voter, than an available alternative. Our real-world data do not come with labels as to when voters are casting misaligned votes, nor which vote is in their best interest. We must posit these ourselves.

We begin by assigning weights to each of the criteria for which we have candidate ratings $\mathbf{r}$. We weight our three dimensions using draws from a Dirichlet distribution, $Dir(\alpha) = \beta$. This gives some common center to each weight offset by random noise. To start, we use $\alpha = (24, 8, 8)$ as our best guess of realistic weights. We explore a wide range of possible weights elsewhere, but report only our best-guess in this paper. Next, we derive the utility-maximizing vote by maximizing the weighted sum of the ratings. For each voter, they would vote sincerely for candidate $c \in C$ if that candidate maximizes their utility:

Principles for fairness metrics

- Be clear about **what** you're trying to measure
- Metric should be **sensitive to what matters**, not sensitive to what doesn't
- For a reasonable regulation, rule should be **achievable** by good-government actors and should include room for exceptional cases
- Induced **incentives** should align with policy goals
- Should admit a healthy **procedure**
- Should be robustly tested against **ground truth** (some instances with agreed labels) and should stand up to sensitivity analysis

Figure 3: Metric Principles

$$vote_{sincere,c} = \begin{cases} 1 & \text{if } \operatorname{argmax}_c[u(c) = \beta^T \mathbf{r}_c] = c, \\ 0 & \text{otherwise} \end{cases}$$

Next, we select some proportion μ of the voters to cast a misaligned vote. These voters vote for a candidate that does not maximize their utility, selected at random from the available options. To begin, we set $\mu = 0.2$. We alter this in [robustness 2].

$$vote_{misaligned,c} = \begin{cases} 1 & \text{if } U(0,1) \geq m \text{ and } vote_{sincere} = 1, \\ 1 & \text{if } U(0,1) < m \text{ and } \operatorname{sample}(C) = c \\ 0 & \text{otherwise} \end{cases}$$

Our first estimand, the misalignment rate, therefore is equal to μ . A diagnostic that returns $\hat{\mu} = \mu$ meets criterion 1.

4.2 System Differences

Achieving criterion 2 is equivalent to minimizing type I error: we do not want false-positives. When there are no differences in the rate of misaligned voting between voting systems, it is important that our diagnostic does not identify one. We test the diagnostic suite against data from

real-world elections under both proportional systems (PR) and single-member district systems (SMD). A diagnostic that satisfies criterion 2 will return $\hat{\mu} = \hat{\mu}_{PR} = \hat{\mu}_{SMD} = m$ when $\mu_{PR} = \mu_{SMD}$. We can observe this in the baseline setup of our simulation.

When a voting system *does* engender more misaligned voting, we want to detect that as well. A good diagnostic, by criterion 3, will minimize type II error. That is, when $\mu_{SMD} = \mu + \varepsilon$, a good test will return both $\hat{\mu} = \mu$ and $\hat{\varepsilon} = \varepsilon$. We test for this by rerunning the simulation but with an increase to the misalignment rate of ε for each voting system, in turn. To start, $\varepsilon = 0.1$, but we vary this in section 5.1.

5 Results

5.1 Baseline

We test many potential diagnostics. Diagnostic A is the current state-of-the-art. It defines $\hat{m}_A$ as the proportion voting for a candidate that doesn't maximize a given dimension. In this and most cases, that dimension is party rating. We identify $\hat{\mu}_A$ first on the sincere votes. In this case, μ , represented by the constant, should equal 0, as should ε . Diagnostic A finds misaligned voting where there is none, and finds more misaligned voting under SMD when, again, there is none. Next we apply this estimator to potentially misaligned votes. While A estimates μ_A well, it misestimates ε , finding a difference between voting systems where there is none.

Estimator B uses the average weights placed on each of the candidate ratings to estimate each candidate's utility and identifies a misaligned vote as any that does not go to the utility estimate-maximizing candidate. If every voter had identical weights, this estimator would be exactly accurate. As it is, we think that this diagnostic represents the best a theory-informed researcher could ever hope to do. Even with knowledge of the average weights and the individual ratings, this estimator still performs poorly. $\hat{\mu}_B = 0.23$ and $\hat{\varepsilon}_B = -0.01$.

Diagnostic C is the Convex Hull approach. It is conservative in identifying cases of misaligned voting and so underestimates μ . $\hat{\mu}_C = 0.169$. However, it is an improvement over the other diagnostics in two respects. First, it does not find a difference between voting systems none exists

	<i>Dependent variable:</i>						
	(Ground Truth)	(A Sincere)	(A Misaligned)	(B Sincere)	(B Misaligned)	(C)	(D)
vote_typeSMD	0.001 (0.002)	0.020*** (0.001)	0.020*** (0.002)	-0.010*** (0.002)	-0.010*** (0.002)	0.002 (0.002)	0.008*** (0.002)
Constant	0.199*** (0.001)	0.015*** (0.001)	0.199*** (0.001)	0.230*** (0.001)	0.230*** (0.001)	0.169*** (0.001)	0.148*** (0.001)
Observations	208,263	208,263	208,263	208,263	208,263	208,263	208,162

Note: misalignment rate $m = 0.2$. $\varepsilon = 0$.

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

Table 1: Baseline

$\hat{\varepsilon} = 0.002$. Second, it cannot, by construction, find misaligned voting when voting is sincere. This is why there is no column (C Sincere). The entries would be 0.

Lastly, diagnostic D is the dominance approach depicted in figure 2. It performs poorly on both $\hat{\mu}$ and $\hat{\varepsilon}$. This is a disappointment.

5.2 Robustness

We have seen how each estimator performs at $\mu = 0.2, \varepsilon = 0$. Now, we extend the analysis to many different combinations of the two parameters. Figure 4 displays 4 plots, one for each diagnostic. Each point represents a different combination of μ and ε . μ ranges from 0 to 0.5 while ε ranges from -0.5 to 0.5. The point (0,0) then, represents totally sincere voting, (0,-.5) represents high misalignment in SMD systems and none in PR, while (0,0.5) represents the opposite. Each point is shaded by how much it misestimates μ , the base-rate. Dark blue is an extreme underestimate, white is accurate, and red is a severe overestimate.

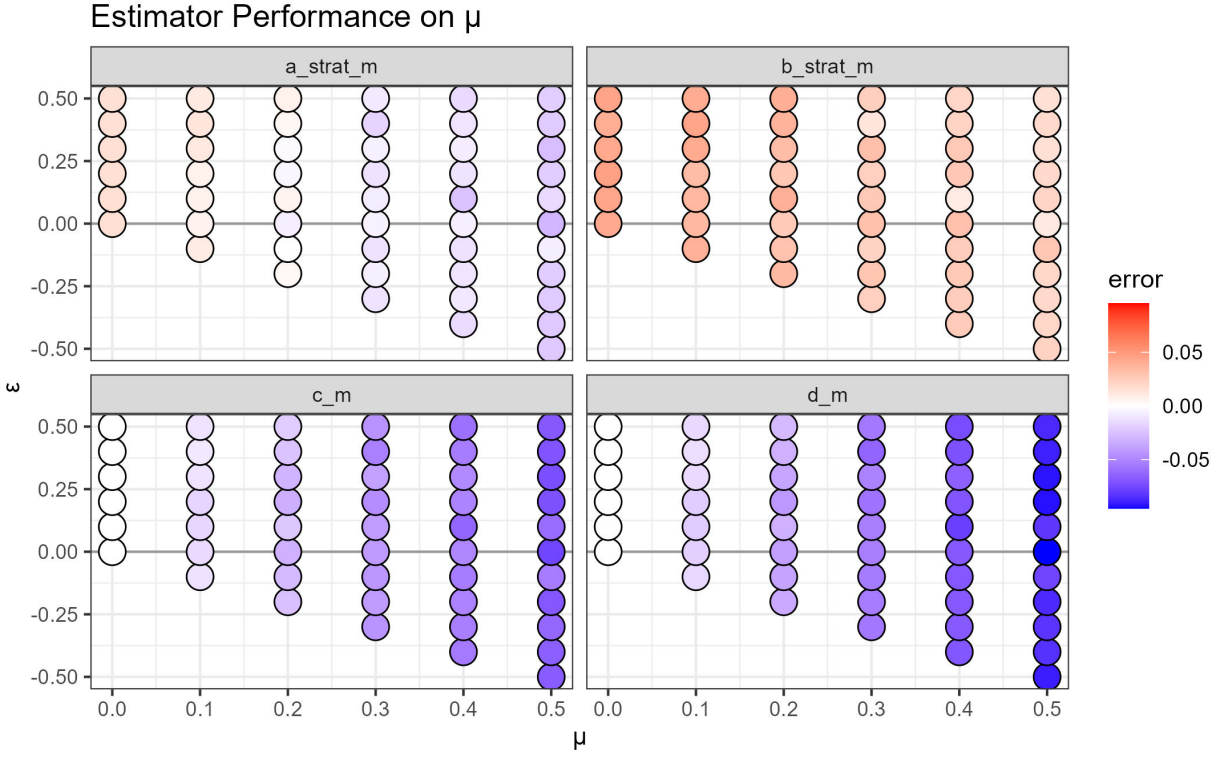


Figure 4: Error on μ

Figure 5 is identical except the shading represents misestimates of ε . As we can see, metrics C and D perform well when $\varepsilon = 0$, and when $\mu = 0$, but are very sensitive to deviations. Their bias, though large, is almost always towards 0. These are conservative estimators of misaligned voting. Metric A, by contrast, has lower error rates in general, but is sometimes biased in the less conservative direction – finding misaligned voting where none exists. In figure 4, any red is bias in this direction. In figure 5, bias away from zero is represented by red above the vertical axis or blue below it. There's work to be done, but I'm happy with this as a start.

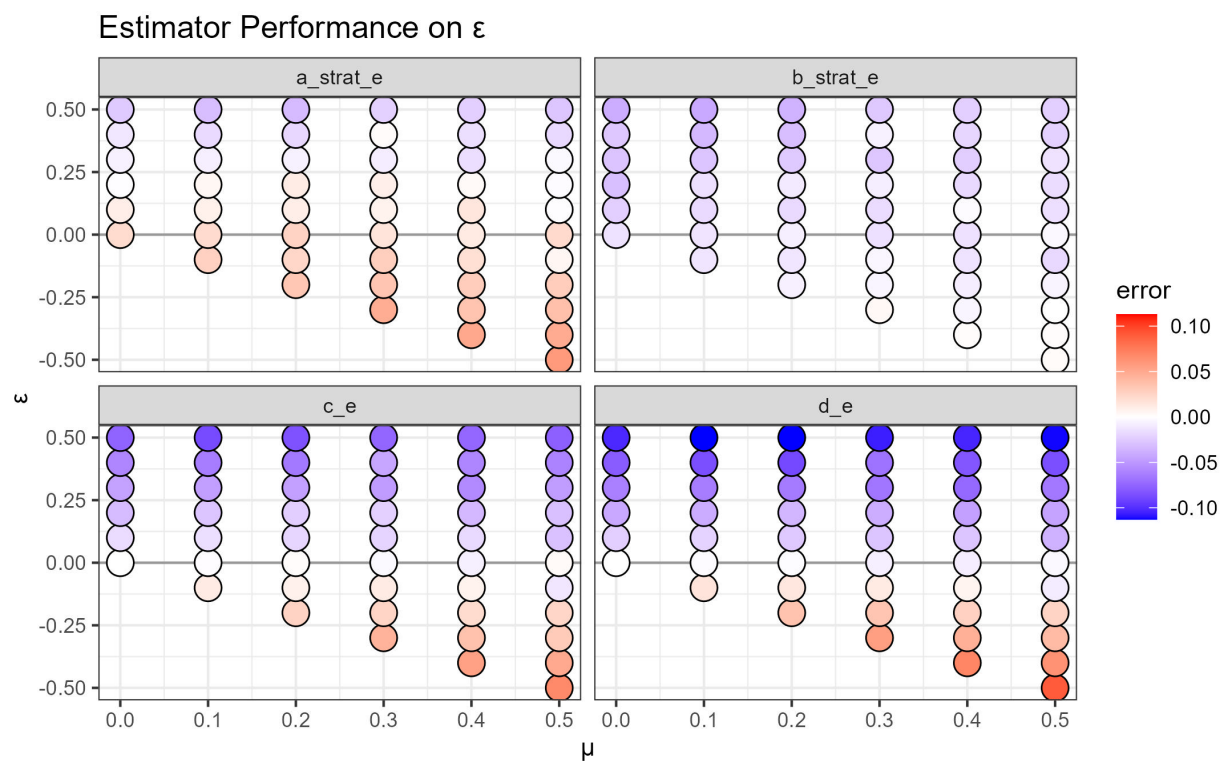


Figure 5: Error on ϵ